

MEDIATEK

MT8676 Yocto OTA User Manual

Version: 1.0
Release date: 2024-09-19

Use of this document and any information contained therein is subject to the terms and conditions set forth in Exhibit 1. This document is subject to change without notice.

Version History

Version	Date	Author	Description
1.0	2024-09-19	Yiru Feng	Official release

Table of Contents

Version History	2
Table of Contents	3
List of Figures	3
List of Tables	3
1 OTA	4
1.1 Overview	4
1.1.1 Brief Introduction	4
1.1.2 Purpose	4
1.1.3 Who Should Read This Document	4
1.1.4 How to Use This Manual	4
1.2 OTA Update Definitions	5
1.3 OTA Abbreviation	5
1.4 Enable A/B System Updates Service	5
1.5 Architecture Overview	5
1.5.1 AB System Updates Process	6
1.5.1.1 How to Use Yocto Full/Delta OTA	7
1.5.2 A/B System Partition Layout	8
1.5.3 Build otapackage Architecture (Temporary Solution)	9
1.5.3.1 Build Yocto targetfiles.zip	9
1.5.3.2 Build Full otapackage.zip	10
1.5.3.3 Build Delta otapackage.zip	10
1.5.4 Change Verification Key for otapackage.zip	11
1.5.5 LK2 Boot Control Flow	12
Exhibit 1 Terms and Conditions	13

List of Figures

Figure 1-1. Yocto OTA update architecture	6
Figure 1-2. Yocto AB partition layout	8
Figure 1-3. Yocto build otapackage architecture	9
Figure 1-4. Yocto LK2 boot flow	12

List of Tables

Table 1-1. Chapter overview	4
Table 1-2. Abbreviations	5

1 OTA

1.1 Overview

1.1.1 Brief Introduction

The introduction to Yocto A/B system update is as follows:

- The system update method is A/B update, where the device runs two completely parallel systems, active_slot and non-active_slot.
- On the disk, always keep a bootable system so that the device will not become bricked.
- During the LK2 phase, the active_slot is confirmed and the corresponding partition data is loaded based on the AB flag set in the misc partition.
- The OTA upgrade runs in the background in normal mode, without user awareness.

1.1.2 Purpose

A guide to AB system updates is provided to users in this document.

The document primarily discusses the methods for generating OTA packages and testing on the Yocto platform; it also covers the utilization of the AB flag in the boot process.

1.1.3 Who Should Read This Document

This document is mainly aimed at the following groups:

- Engineers with knowledge of OTA A/B system updates (seamless updates)

1.1.4 How to Use This Manual

The distribution of information within this document is explained in this section, along with the provision of tips and examples to facilitate locating and comprehending information within this document.

[Table 1-1](#) outlines the chapters in this document.

Table 1-1. Chapter overview

No.	Chapter	Contents
1.1	Introduction	Described the scope and layout of this document
1.2	OTA Update Definitions	Definition of OTA upgrade
1.3	Abbreviations	OTA abbreviation
1.4	Enable the A/B System Updates Service	Open the required service for AB system upgrade on the MediaTek platform.
1.5.1	A/B System Update Process	Introduction to the architecture and upgrade methods of the AB system
1.5.2	A/B System Partition Layout	Introduction to A/B partitioning

No.	Chapter	Contents
1.5.3	Build otapackage Architecture	Introduce how to create OTA full upgrade package and differential upgrade package
1.5.4	Change Verification Key for otapackage.zip	Introduce the upgrade package signing key and its customization
1.5.5	LK2 Boot Control Flow	Introduction to bootctl parameters in Yocto LK2 phase

1.2 OTA Update Definitions

The professional terminology of OTA is defined as follows:

Full Update: A full update refers to the complete update of the partition data to be upgraded on the device, with the update package containing complete image information for each partition.

Delta Update: A delta update, also known as incremental update, refers to updating the differential part of the device's partition data to the corresponding partition data of the target version. The upgrade package only contains the differential parts of two versions, which can greatly reduce the size of the upgrade package.

Note: Only install the corresponding incremental update package on old or source devices used when compiling differential packages.

1.3 OTA Abbreviation

Table 1-2. Abbreviations

Abbreviations	Explanation
Bootctl	Boot control, A/B slot identification structure, stored in the misc partition
LK2	Little Kernel 2, Yocto boot stage
OTA	Over-The-Air, divided into full updates and delta updates.

1.4 Enable A/B System Updates Service

Please add the following module to the corresponding `project.bb` file:

- `recipes-auto/images/mtk-core-image-auto8676.bb:`
 - `update-engine-sideload-u \`
 - `update-verifier-u \`

update-engine-sideload-u: OTA upgrade module, perform partition read and write operations

update-verifier-u: After successful OTA reboot, modify the new slot flag

1.5 Architecture Overview

This section first briefly describes the various modules of the system and their relationships:

1. A/B system update process
2. A/B system partition layout
3. Build otapackage architecture
4. Change the verification key for otapackage.zip
5. LK2 boot control flow

1.5.1 AB System Updates Process

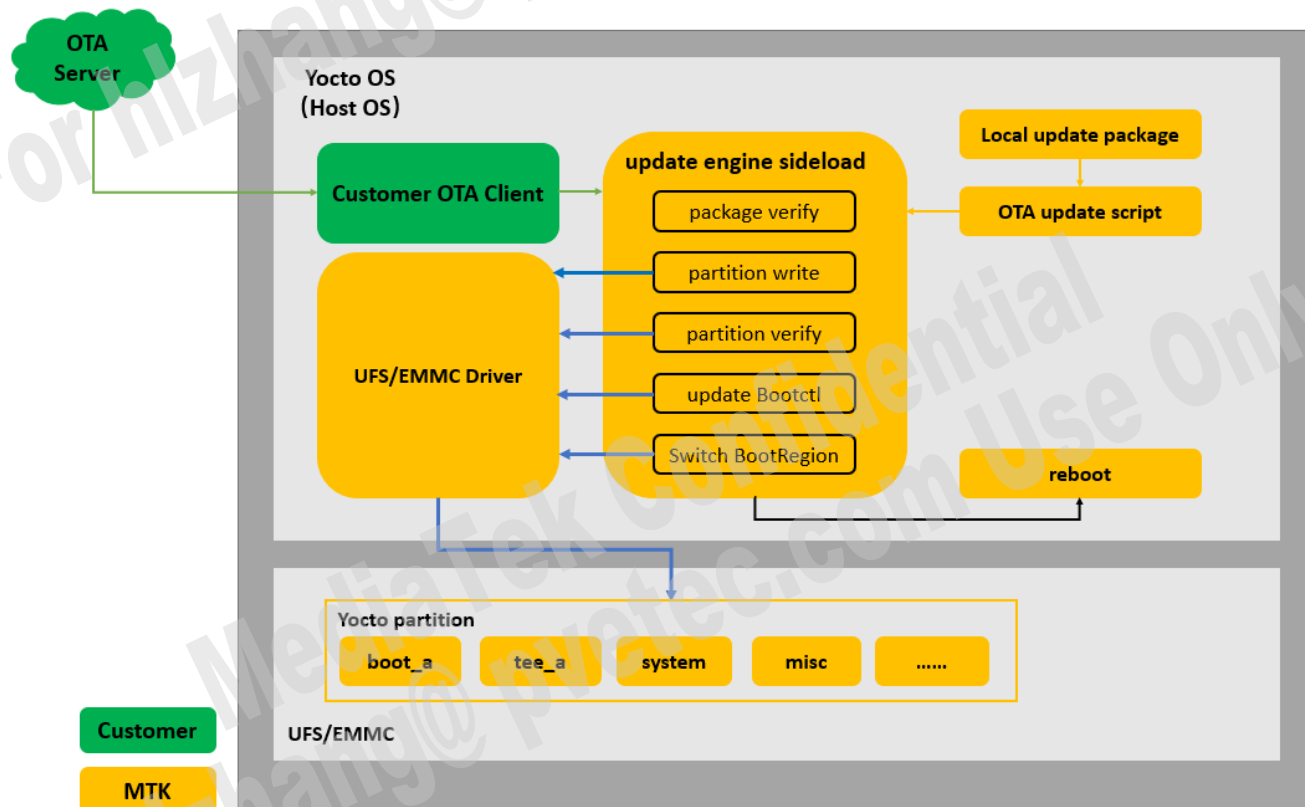


Figure 1-1. Yocto OTA update architecture

Figure 1-1 depicts the overall architecture of Yocto OTA update, from which the following information can be obtained:

1. Yocto OTA updates can be divided into two steps: get otapackage and trigger process.
MediaTek only provides OTA capabilities. otapackage is stored in the local environment and uses update script triggering the process.
Customers can store otapackage in remote server or storage devices and implement triggering methods like APP themselves.
2. Taking local update as an example, the general flow of OTA update is as follows:
 - Store the update package in the local PC environment.
 - Executing the OTA update script calls update_engine sideload to trigger the update.
 - **Verify the integrity and accuracy of the update package (refer to Section 1.5.4 for key verification details)**
 - Update the update data in the update package to the corresponding partitions respectively.
 - Validate the data that has been updated in the partition to ensure data accuracy.

- Update the system boot control parameters, i.e. Bootctl information, to ensure that the system loads new slot partition data at the next boot.
- Switch boot region to ensure that the system loads new preloader partition data at the next startup.
- Reboot the device.

1.5.1.1 How to Use Yocto Full/Delta OTA

Test Environment Setup

1. The PC has a Python 3 environment.
2. Flash the base load in the DUT.
3. Get OTA package.

Test Step

1. Device connected to Yocto ADB environment
2. Execute the update script: `python3 hypervisor_update.py --file otapackage_delta.zip > update.txt 2>&1`

Note:

- `hypervisor_update.py`: Update script
- `otapackage_delta.zip`: OTA package, unlimited package name
- `> update.txt 2>&1`: Store the upgrade log in `update.txt` in the current directory

Expected result

1. Update should be completed, and `update.txt` prints the following log: [INFO:update_attempter_android.cc(600)]
Update successfully applied, waiting to reboot.
2. The first restart after the upgrade should successfully enter the home screen.
3. Check whether the setting version has been updated.

Note: If the test upgrade fails, please provide the update log and UART log in CR.

1.5.2 A/B System Partition Layout

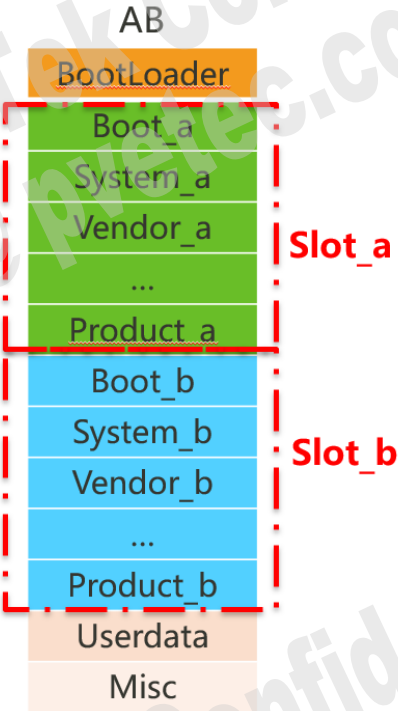


Figure 1-2. Yocto AB partition layout

Yocto AB partition layout is as shown in Figure 1-2, and AB partition configuration can be done in the partition table: meta/meta-mediatek-mt8676/recipes-bsp/ptgen/files/auto8676p1_64/partition_table_emmc_ab.csv

To reduce the update of AB partition, the target partition must first be modified to a single partition, and the entry in the partition table corresponding to “OTA_Update” should be changed to “N”.

Partition_Name,Type,Size_KB,Region,,Reserved,Download,Download_File,OTA_Update,EmptyBoot_Needed,Oper	FILE
1 Partition_Name,Type,Size_KB,Region,,Reserved,Download,Download_File,OTA_Update,EmptyBoot_Needed,Oper	1 Partition_Name,Type,Size_KB,Region,,Reserved,Download,Download_File,OTA_Update,EmptyBoot_Needed,Oper
2 ...emmc,ufs,.....	2 ...emmc,ufs,.....
3 preloader_a,Raw data,1024,EMMC_BOOT1,UFS_LU0,N,Y,b12.img,Y,N,BOOTLOADERS,Y	3 preloader_a,Raw data,1024,EMMC_BOOT1,UFS_LU0,N,Y,b12.img,N,N,BOOTLOADERS,Y
4 preloader_b,Raw data,1024,EMMC_BOOT2,UFS_LU1,N,Y,b12.img,Y,N,BOOTLOADERS,Y	4 preloader_b,Raw data,1024,EMMC_BOOT2,UFS_LU1,N,Y,b12.img,N,N,BOOTLOADERS,Y
5 pgpt,Raw data,32,EMMC_USER,UFS_LU2,N,N,NONE,N,N,AUTO,Y	5 pgpt,Raw data,32,EMMC_USER,UFS_LU2,N,N,NONE,N,N,AUTO,Y
6 misc,Raw data,512,EMMC_USER,UFS_LU2,N,N,NONE,N,N,AUTO,Y	6 misc,Raw data,512,EMMC_USER,UFS_LU2,N,N,NONE,N,N,AUTO,Y
7 vb,Raw data,4096,EMMC_USER,UFS_LU2,N,N,NONE,N,N,AUTO,Y	7 vb,Raw data,4096,EMMC_USER,UFS_LU2,N,N,NONE,N,N,AUTO,Y
8 para,Raw data,512,EMMC_USER,UFS_LU2,N,N,NONE,N,N,AUTO,Y	8 para,Raw data,512,EMMC_USER,UFS_LU2,N,N,NONE,N,N,AUTO,Y
9 expdb,Raw data,131072,EMMC_USER,UFS_LU2,N,N,NONE,N,N,AUTO,Y	9 expdb,Raw data,131072,EMMC_USER,UFS_LU2,N,N,NONE,N,N,AUTO,Y
10 frp,Raw data,1024,EMMC_USER,UFS_LU2,N,N,NONE,N,N,AUTO,Y	10 frp,Raw data,1024,EMMC_USER,UFS_LU2,N,N,NONE,N,N,AUTO,Y
11 nvfcg,EXT4,32768,EMMC_USER,UFS_LU2,N,N,NONE,N,N,PROTECTED,Y	11 nvfcg,EXT4,32768,EMMC_USER,UFS_LU2,N,N,NONE,N,N,PROTECTED,Y
12 nvdata,EXT4,81920,EMMC_USER,UFS_LU2,N,Y,nvdata.ext4,N,N,AUTO,Y	12 nvdata,EXT4,81920,EMMC_USER,UFS_LU2,N,Y,nvdata.ext4,N,N,AUTO,Y
13 md_sec,EXT4,8192,EMMC_USER,UFS_LU2,N,N,NONE,N,N,AUTO,Y	13 md_sec,EXT4,8192,EMMC_USER,UFS_LU2,N,N,NONE,N,N,AUTO,Y
14 vbmeta_a,Raw data,8192,EMMC_USER,UFS_LU2,N,Y,vbmeta.img,Y,N,AUTO,Y	14 vbmeta_a,Raw data,8192,EMMC_USER,UFS_LU2,N,Y,vbmeta.img,Y,N,AUTO,Y

For the update of AB partitions, it is necessary to simultaneously create two new partitions, partition_a and partition_b, and set the corresponding “OTA_Update” field in the partition table to “Y”. The size of AB partitions must remain consistent.

1.5.3 Build otapackage Architecture (Temporary Solution)

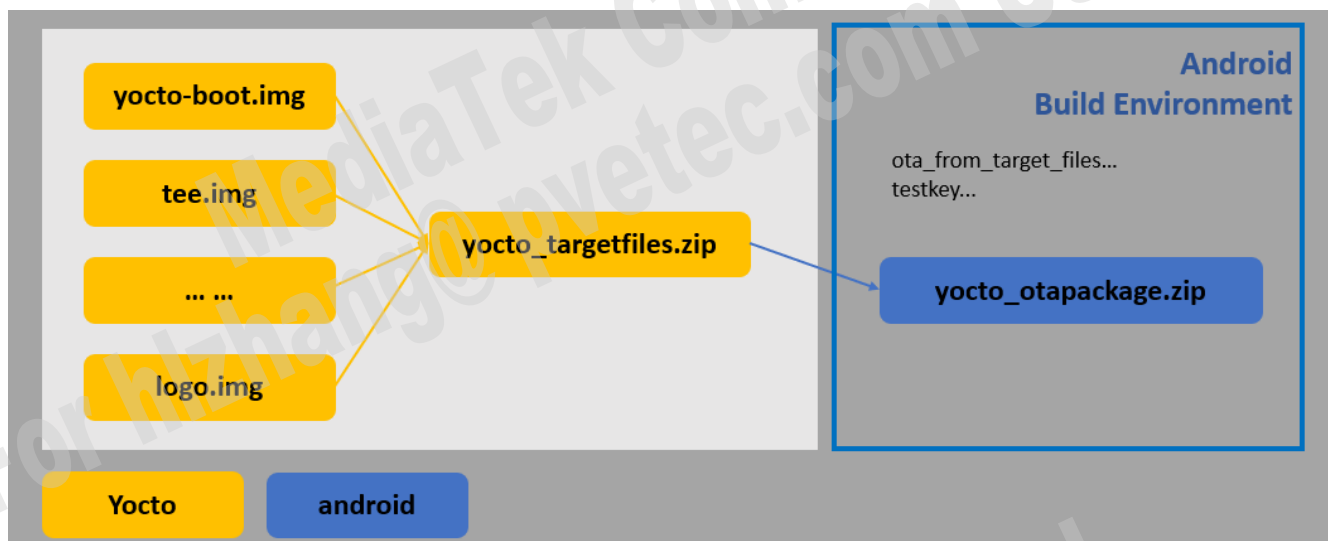


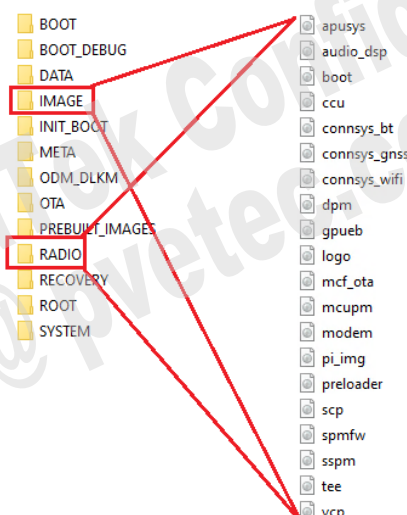
Figure 1-3. Yocto build otapackage architecture

Figure 1-3 describes the architecture of the Yocto OTA compilation update package, from which the following information can be obtained:

1. Compiling the Yocto update package temporarily needs to be done using Android's compilation environment
2. Yocto Only compiles the update package roughly following the flow below:
 - Compile Yocto targetfiles.zip
 - Using the Yocto targetfiles.zip, compile and generate full update packages and differential update packages in the Android compilation environment.

1.5.3.1 Build Yocto targetfiles.zip

The Yocto targetfiles.zip **needs manual replacement of all AB images of the new version** to /default_target/IMAGE and /RADIO, then rezip to generate yocto_target.zip.



Execute the command “`zip -r yocto_target.zip .`” in the current `/default_target/` directory.

1.5.3.2 Build Full otapackage.zip

After obtaining `yocto_target.zip`, the full update package for Yocto can be compiled in the Android build environment.

Compilation instructions:

```
./out_sys/host/linux-x86/bin/ota_from_target_files -v --block -p out_sys/host/linux-x86 -k
build/make/target/product/security/testkey --skip_postinstall yocto_target.zip.zip
otapackage_full.zip
```

Note:

- `ota_from_target_files`: Update package compilation script
- `build/make/target/product/security/testkey`: The key path for signing the update package. MediaTek defaults to using the testkey provided by Google, but your company can specify a different key path using the `-k` parameter. To change the verification key, please refer to [Section 1.5.4](#).
- `yocto_target.zip`: Yocto target files.zip
- `otapackage_full.zip`: The final Yocto full update package, the package name can be freely specified.
- Please make sure to compile the update package in the Android build environment and ensure that “source & lunch” is executed before compiling the entire package.

1.5.3.3 Build Delta otapackage.zip

Prerequisite:

To compile a differential package, it is necessary to prepare two copies of `yocto_target.zip` in advance. One is the base version (source version) of `yocto_target.zip`, named as `source_yocto_target.zip` here; the other is the target version of `yocto_target.zip`, named as `target_yocto_target.zip` here.

Note: Please ensure that the base version load burned on the platform and the base version `target_files.zip` are compiled from the same build.

Compilation instruction:

```
./out_sys/host/linux-x86/bin/ota_from_target_files -v --block -p out_sys/host/linux-x86 -k
build/make/target/product/security/testkey --skip_postinstall -i
source_yocto_target_files.zip target_yocto_target_files.zip otapackage_delta.zip
```

Note:

- ota_from_target_files: Update package compilation script
- build/make/target/product/security/testkey: The key path for signing the update package. MediaTek defaults to using the testkey provided by Google. Your company can specify a different key path by using the `-k` parameter. To change the signature key, please refer to [Section 1.5.4](#).
- source_yocto_target_files.zip: yocto_target.zip for the source version
- target_yocto_target_files.zip: yocto_target.zip for the target version
- otapackage_delta.zip: The final Yocto differential update package, the package name can be freely specified
- Please make sure to compile the update package in the Android build environment and ensure that “source & lunch” is executed before compiling the entire package.

1.5.4 Change Verification Key for otapackage.zip

OTA update will use two keys: **xxx.pk8** and **xxx.x509.pem**.

1. The key in xxx.pk8 format is used for signing the update package during packaging.
2. The key in xxx.x509.pem format is used to verify the update package during the OTA update process.

Please refer to the following two steps to modify the signature key of the update package:

1. Replace the compilation instructions in [1.5.3.2](#) for compiling the full package and in [1.5.3.3](#) for compiling the differential package after changing the `-k` parameter to use the paths of xxx.pk8 and xxx.x509.pem files.
2. Replace the actual xxx.x509.pem file used inside otacerts.zip located at meta/meta-MediaTek/recipes-support/update-engine-sideload-u/files/ota.

After the key is replaced, it is necessary to confirm that the following two files are consistent before the update verification can be passed.

- /META-INF/com/android/otacert file for otapackage.zip file
- /system/etc/security/otacerts.zip file on the device

1.5.5 LK2 Boot Control Flow

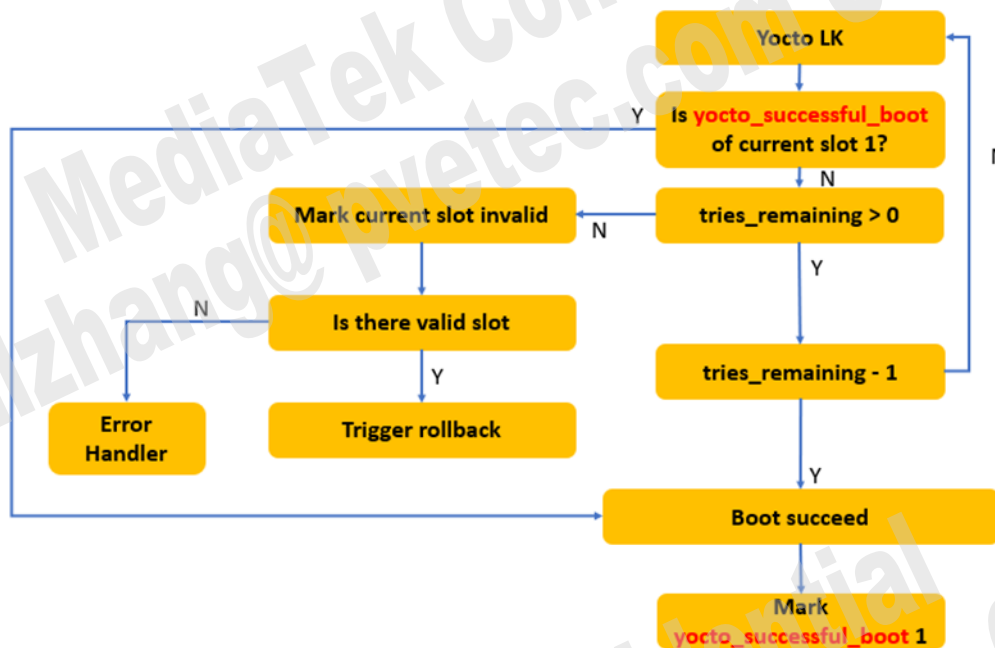


Figure 1-4. Yocto LK2 boot flow

1. Upon completion of the OTA update, a restart is initiated, leading to the startup of Yocto OS and entering the Yocto LK2 phase.
2. Check if Yocto has been successfully booted in the current slot by verifying if the value of “Yocto successful boot” in the boot control is 1.
3. If a value of 1 is assigned to “Yocto successful boot”, it indicates that the current slot is “bootable”, no action should be taken, and the machine will proceed with booting. In case “Yocto successful boot” holds a value of 0, indicating that the current slot has not booted successfully previously, then verify if the restart count for the current slot exceeds 0 by examining tries_remaining within boot control.
4. If the value of tries_remaining is above 0, decrement it by 1 and proceed with the boot process; otherwise, indicating that the current slot is “not bootable”, mark it as invalid by setting both its priority and “Yocto successful boot” to 0.
5. Next, the current availability of slots is checked to determine if there are any with a priority other than 0. If none are found, an error message is generated by the system. If available, the priority of the rollback slot is set to 15 and Yocto is triggered to restart for completing the rollback.
6. When Yocto successfully boots, set the value of “Yocto successful boot” for the current slot to 1.

Exhibit 1 Terms and Conditions

Your access to and use of this document and the information contained herein (collectively this “Document”) is subject to your (including the corporation or other legal entity you represent, collectively “You”) acceptance of the terms and conditions set forth below (“T&C”). By using, accessing or downloading this Document, You are accepting the T&C and agree to be bound by the T&C. If You don’t agree to the T&C, You may not use this Document and shall immediately destroy any copy thereof.

This Document contains information that is confidential and proprietary to MediaTek Inc. and/or its affiliates (collectively “MediaTek”) or its licensors and is provided solely for Your internal use with MediaTek’s chipset(s) described in this Document and shall not be used for any other purposes (including but not limited to identifying or providing evidence to support any potential patent infringement claim against MediaTek or any of MediaTek’s suppliers and/or direct or indirect customers). Unauthorized use or disclosure of the information contained herein is prohibited. You agree to indemnify MediaTek for any loss or damages suffered by MediaTek for Your unauthorized use or disclosure of this Document, in whole or in part.

MediaTek and its licensors retain titles and all ownership rights in and to this Document and no license (express or implied, by estoppels or otherwise) to any intellectual propriety rights is granted hereunder. This Document is subject to change without further notification. MediaTek does not assume any responsibility arising out of or in connection with any use of, or reliance on, this Document, and specifically disclaims any and all liability, including, without limitation, consequential or incidental damages.

THIS DOCUMENT AND ANY OTHER MATERIALS OR TECHNICAL SUPPORT PROVIDED BY MEDIATEK IN CONNECTION WITH THIS DOCUMENT, IF ANY, ARE PROVIDED “AS IS” WITHOUT WARRANTY OF ANY KIND, WHETHER EXPRESS, IMPLIED, STATUTORY, OR OTHERWISE. MEDIATEK SPECIFICALLY DISCLAIMS ALL WARRANTIES OF MERCHANTABILITY, NON-INFRINGEMENT, FITNESS FOR A PARTICULAR PURPOSE, COMPLETENESS OR ACCURACY AND ALL WARRANTIES ARISING OUT OF TRADE USAGE OR OUT OF A COURSE OF DEALING OR COURSE OF PERFORMANCE. MEDIATEK SHALL NOT BE RESPONSIBLE FOR ANY MEDIATEK DELIVERABLES MADE TO MEET YOUR SPECIFICATIONS OR TO CONFORM TO A PARTICULAR STANDARD OR OPEN FORUM.

Without limiting the generality of the foregoing, MediaTek makes no warranty, representation or guarantee regarding the suitability of its products for any particular purpose, nor does MediaTek assume any liability arising out of the application or use of any product, circuit or software. You agree that You are solely responsible for the designing, validating and testing Your product incorporating MediaTek’s product and ensure such product meets applicable standards and any safety, security or other requirements.

The above T&C and all acts in connection with the T&C or this Document shall be governed, construed and interpreted in accordance with the laws of Taiwan, without giving effect to the principles of conflicts of law.